
7 Summarise Literature

New Description

Outcome	Weight
Context of Research Project	◆◆◆◆

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Outcome	Weight
Communication of Research Outcomes	◆◆◆◆◇

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Outcome	Weight
Validation of Objectives and Methodology	◆◆◆◆

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Outcome	Weight
Application of Skills and Knowledge	◆◆◆◆

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Outcome	Weight
Explore of IT Areas	◆◆◆◆

Date	Author	Comment
2025/08/30 01:50	N/a Nikhil	Ready to Mark
2025/08/30 08:39	Ananda Maiti	Complete

DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Summarise Literature

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2025/08/30 01:50

Tutor:
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Outcome	Weight
Context of Research Project	◆◆◆◆◆
Communication of Research Outcomes	◆◆◆◆◇
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Application of Skills and Knowledge	◆◆◆◆◆

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August 30, 2025





Literature Review Summary

GPS Spoofing Detection and Countermeasures

Analysis of Research Papers on GNSS Spoofing Vulnerabilities

Student Details

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University: Deakin University
Course: SIT723 Research Techniques & Applications
SIT792 Minor Thesis
Task: 2.1P - Summarise Literature
Date: August 30, 2025

Task Objectives

- Find relevant literature to understand GPS spoofing in autonomous systems
- Read and summarise at least 5 research papers
- Extract relevant information for project development
- Demonstrate research and analysis capabilities

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1 Executive Summary

This literature review examines GPS spoofing detection methodologies for autonomous systems, focusing on unmanned aerial vehicles (UAVs) and mobile robots. The analysis covers five key research papers that address different aspects of GNSS spoofing vulnerabilities, detection techniques, and countermeasures. The review identifies current research gaps, technological challenges, and future directions for securing autonomous navigation systems against GPS spoofing attacks.

2 Literature Analysis

2.1 Paper 1: Analysis of GPS Spoofing Vulnerability in Drone 3DR Solo

Authors: Sandra Perez Arteaga et al.

Focus: Commercial drone vulnerability assessment

Key Findings:

- Demonstrated successful GPS spoofing on 3DR Solo drone
- Identified lack of robust security in commercial UAVs
- Highlighted need for enhanced GPS authentication

2.2 Paper 2: Impact and Detection of GPS Spoofing and Countermeasures

Authors: Mukhtar Ahmad et al.

Focus: Comprehensive spoofing detection techniques

Key Contributions:

- Classification of spoofing generation techniques
- Analysis of power monitoring methods (C/N0, AGC)
- Spatial processing approaches using antenna arrays

2.3 Paper 3: Research Progress of GNSS Spoofing Detection Technology

Authors: Li Junzhi et al.

Focus: Systematic review of detection technologies

Research Gaps Identified:

- Need for universal detection methods
- Challenges in complex environments
- Integration of generative adversarial networks (GANs)

2.4 Paper 4: Sensor Fusion-Based GNSS Spoofing Attack Detection

Authors: Sagar Dasgupta et al.

Focus: Multi-sensor fusion approach for autonomous vehicles

Novel Contributions:

- LSTM-based location shift prediction
- k-NN and DTW algorithms for turn detection
- Evaluation on Honda Research Institute Dataset

2.5 Paper 5: Signal Characteristic Analysis and Anomaly Detection

Authors: Usman Tariq, Bilal Tariq

Focus: Hybrid machine learning approach for VANETs

Key Methodologies:

- Bayesian Networks combined with Watchdog Model
- Zero-Knowledge Proofs for authentication
- Multi-layered defense mechanism

3 Comparative Analysis

Paper	Detection Method	Platform	Advantages/Limitations
3DR Solo Analysis	Practical vulnerability testing	Commercial drone	Real-world validation / Limited scope
Ahmad et al.	Signal power monitoring	General GNSS	Comprehensive / Hardware dependent
Li Junzhi et al.	Systematic review	Multiple platforms	Broad coverage / Theoretical focus

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Paper	Detection Method	Platform	Advantages/Limitations
Dasgupta et al.	Sensor fusion + ML	Autonomous vehicles	Novel approach / Dataset limitations
Tariq & Tariq	Hybrid ML (BN+WM)	VANETs	Advanced ML / Complexity

4 Key Research Insights

4.1 Technical Approaches

- Signal-level analysis (power, correlation)
- Spatial processing with antenna arrays
- Machine learning classification
- Multi-sensor fusion strategies
- Cryptographic authentication

4.2 Common Challenges

- Real-time processing requirements
- Sophisticated attack resilience
- Computational resource constraints
- False positive minimization
- Integration complexity

5 Identified Research Gaps

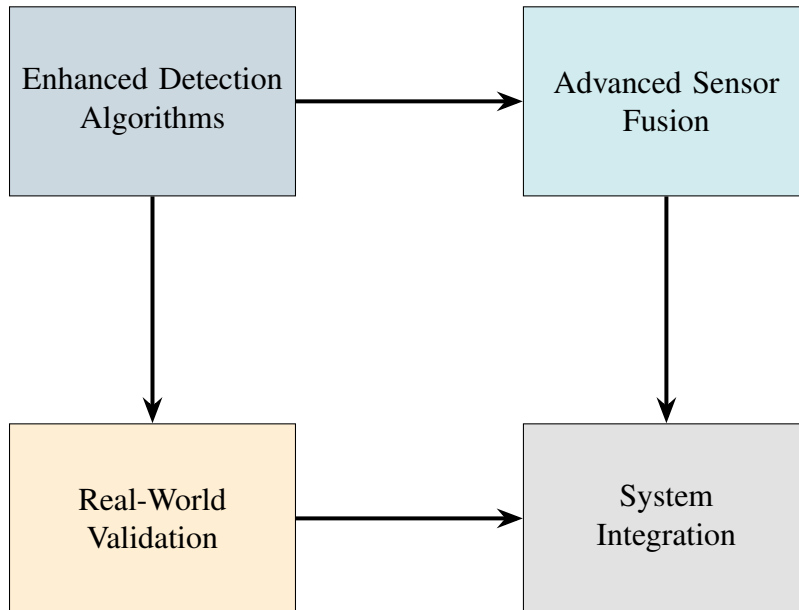
5.1 Technical Gaps

1. **Lightweight Detection Algorithms:** Need for computationally efficient methods suitable for embedded systems
2. **Sophisticated Attack Resistance:** Limited evaluation against advanced, synchronized spoofing
3. **Multi-Platform Validation:** Insufficient testing across diverse autonomous platforms
4. **Real-Time Integration:** Challenges in seamless integration with existing navigation systems

5.2 Implementation Challenges

1. **Standardization:** Lack of unified evaluation metrics and datasets
2. **Scalability:** Limited analysis of coordinated, multi-vehicle attacks
3. **Environmental Adaptability:** Performance degradation in challenging conditions
4. **Cost-Effectiveness:** Balance between security and implementation costs

6 Implications for Future Research



6.1 Practical Applications

- Development of ROS/PX4 integrated detection frameworks
- Creation of standardized spoofing attack datasets
- Implementation of dynamic trust assessment systems
- Design of fail-safe navigation protocols

7 Conclusion

The literature review reveals significant progress in GPS spoofing detection research, with diverse approaches ranging from signal-level analysis to advanced machine learning techniques. However, substantial gaps remain in real-world validation, computational efficiency, and integration with existing autonomous systems.

Future research should focus on developing lightweight, adaptive algorithms that can operate effectively on resource-constrained platforms while maintaining high detection accuracy against sophisticated attacks. The integration of multiple detection strategies in a unified framework represents a promising direction for enhancing the security of autonomous navigation systems.

This analysis provides a solid foundation for understanding current capabilities and limitations in GPS spoofing detection, directly supporting the development of more robust and practical defense mechanisms for autonomous vehicles and mobile robots.

8 References

References

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- [5] Tariq, U., & Tariq, B. (2025). Signal Characteristic Analysis and Anomaly Detection for GPS Spoofing Mitigation. *Ubiquitous Technology Journal*, 1(1), 10-22.